Growth hormone, functions synthesis and related disorders

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DIKIOHS,DUHS

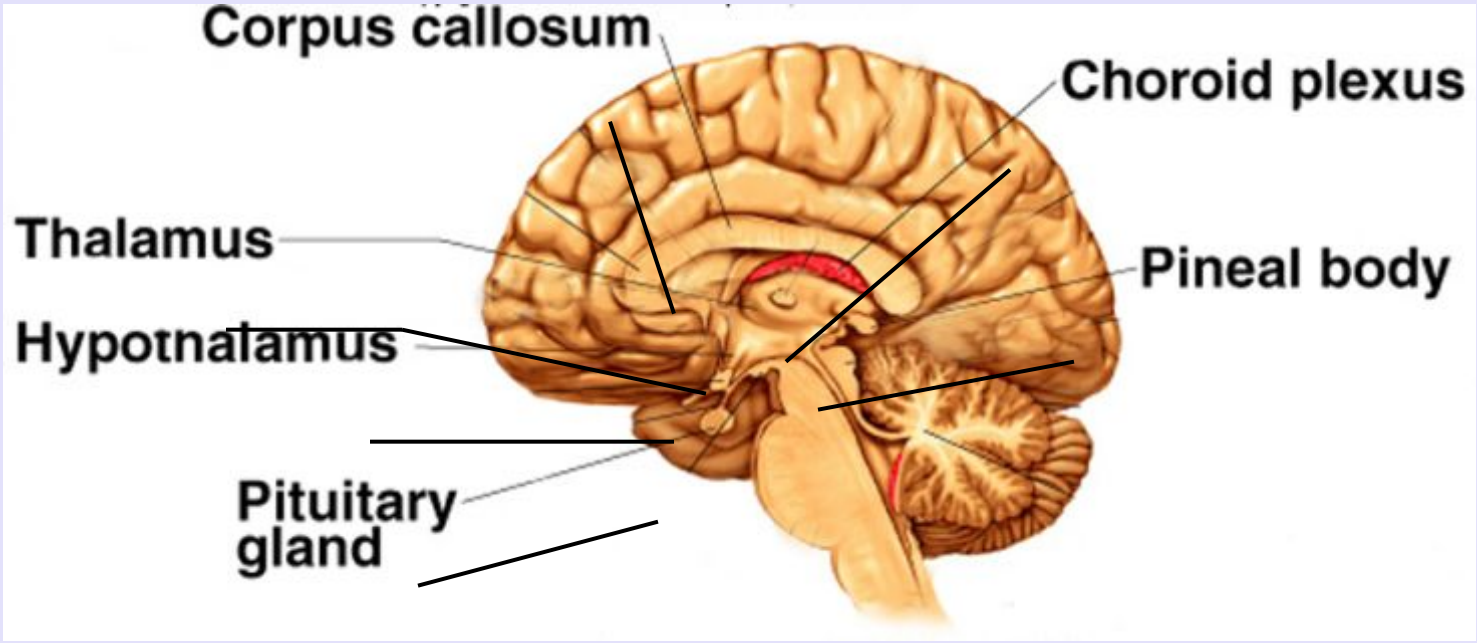
Learning objectives

At the end of lecture student should be able to

- Describe Endocrine control of growth.
- Explain Growth promoting and metabolic functions of growth hormone.
- Discuss Growth hormone and role of somatomedins.
- Enumerate factors influencing the growth hormone secretion.
- Describe the regulation of synthesis of growth hormone.
- Discuss the disorders related with GH excess and deficiency.

Pituitary Gland

- Pituitary gland or hypophysis is a small endocrine gland with a diameter of 1 cm and weight of 0.5 to 1 g.
- It is situated in a depression called ‘sella turcica’, present in the sphenoid bone at the base of skull.
- It is connected with the hypothalamus by the pituitary stalk or hypophyseal stalk.



Divisions Of Pituitary Gland (hypophysis)

- Pituitary gland is divided into two divisions:

1. Anterior pituitary or adenohypophysis

2. Posterior pituitary or neurohypophysis.

- Both the divisions are situated close to each other.

Still both are entirely different in their development, structure and function.

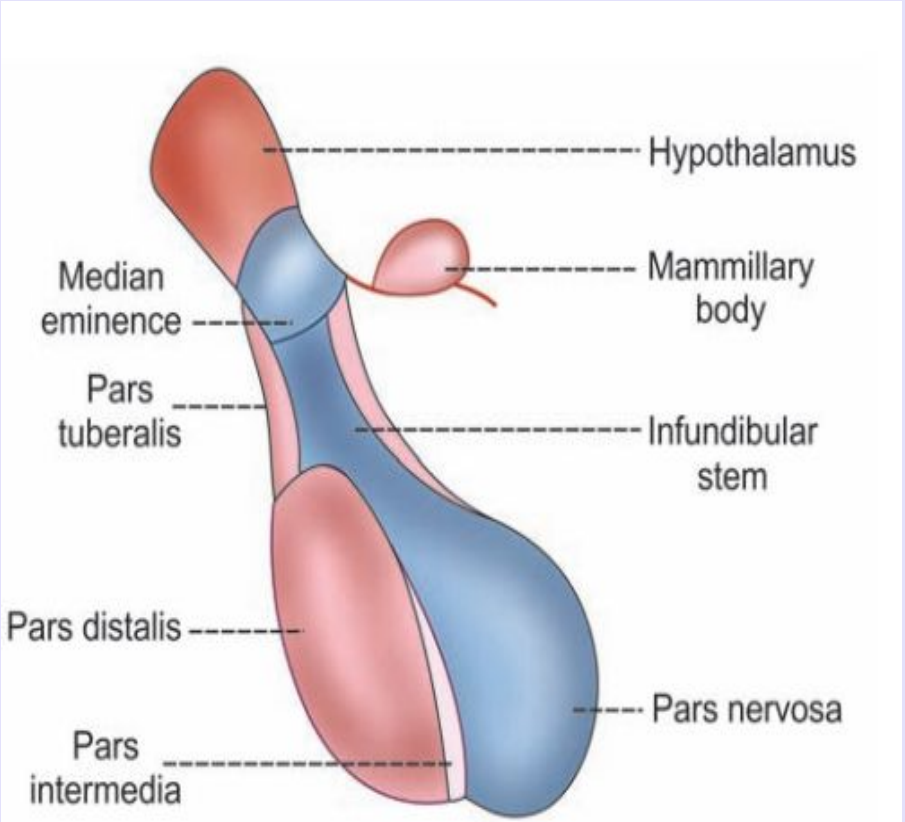


FIGURE 66.1: Parts of pituitary gland
Adenohypophysis Neurohypophysis

- Embryologically the two parts develop from different sources.
- Anterior pituitary is ectodermal in origin and arises from the pharyngeal epithelium as an upward growth known as Rathke pouch.
- Posterior pituitary is neuroectodermal in origin and arises from hypothalamus as a downward diverticulum.
- Rathke pouch and the downward diverticulum from hypothalamus grow towards each other and meet in the midway between the roof of the buccal cavity and base of brain. There, the two structures lie close together.
- Anterior produces own hormones, controlled by hypothalamus.
- Posterior stores & releases hormones made in hypothalamus.

Hypothalamo-hypophyseal Relationship

- The relationship between hypothalamus and pituitary gland is called hypothalamo-hypophyseal relationship.
- Hormones secreted by hypothalamus are transported to anterior pituitary and posterior pituitary. But the mode of transport of these hormones is different.
- Hormones from hypothalamus are transported to anterior pituitary through hypothalamo-hypophysial portal blood vessels.
- But, the hormones from hypothalamus to posterior pituitary are transported by nerve fibers of hypothalamo hypophyseal tract.

- Hypothalamus controls pituitary secretion
- There are neural connections between the hypothalamus and the posterior pituitary and vascular connections between the hypothalamus and anterior pituitary.These hypothalamic releasing and inhibitory hormones are conducted through minute blood vessels called hypothalamic-hypophyseal portal vessels to the anterior pituitary gland.
- The hypothalamic-hypophyseal portal vessels form a direct vascular link between hypothalamus and the anterior pituitary.

HORMONES SECRETED BY ANTERIOR PITUITARY

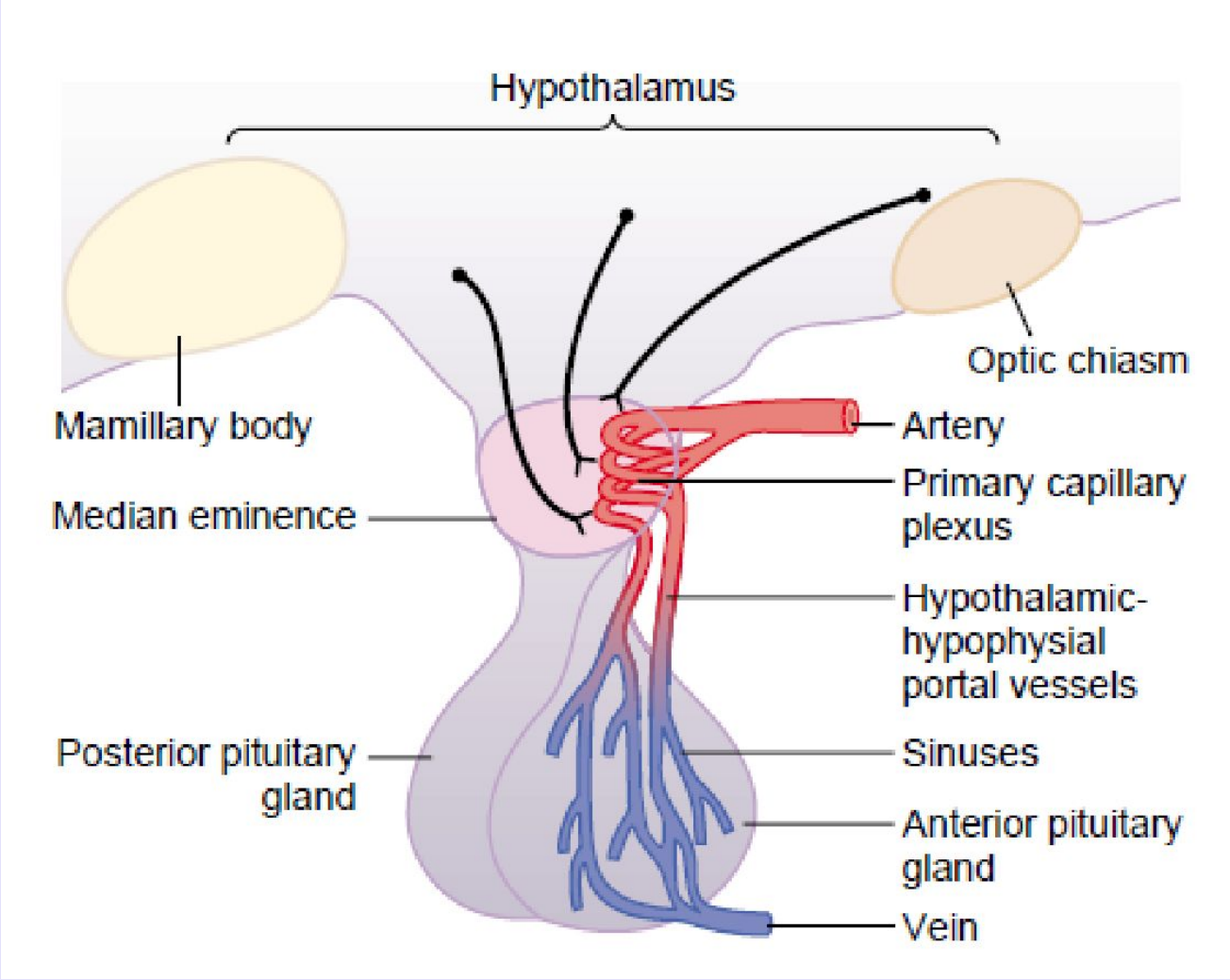
- Six hormones are secreted by the anterior pituitary:

1. Growth hormone (GH) or somatotropic hormone(STH)
2. Thyroid-stimulating hormone (TSH) or thyrotropic hormone
3. Adrenocorticotropic hormone (ACTH)
4. Follicle-stimulating hormone (FSH)
5. Luteinizing hormone (LH) in females or interstitial cell-stimulating hormone (ICSH) in males
6. Prolactin.

Hormones Of Anterior Pituitary

Gland of Origin	Hormones*	Chemical Classification†	Major Actions
Hypothalamus	Thyrotropin-releasing hormone (TRH)	Peptide	Stimulates secretion of TSH and prolactin
	Corticotropin-releasing hormone (CRH)	Peptide	Stimulates secretion of ACTH
	Gonadotropin-releasing hormone (GnRH)	Peptide	Stimulates secretion of LH and FSH
	Somatostatin or somatotropin release–inhibiting hormone (SRIF)	Peptide	Inhibits secretion of growth hormone
	Dopamine or prolactin-inhibiting factor (PIF)	Amine	Inhibits secretion of prolactin
	Growth hormone–releasing hormone (GHRH)	Peptide	Stimulates secretion of growth hormone
Anterior Pituitary	Thyroid-stimulating hormone (TSH)	Peptide	Stimulates synthesis and secretion of thyroid hormones
	Follicle-stimulating hormone (FSH)	Peptide	Stimulates sperm maturation in Sertoli cells of testes
			Stimulates follicular development and estrogen synthesis in ovaries
	Luteinizing hormone (LH)	Peptide	Stimulates testosterone synthesis in Leydig cells of testes
			Stimulates ovulation, formation of corpus luteum, estrogen and progesterone synthesis in ovaries
	Growth hormone	Peptide	Stimulates protein synthesis and overall growth
	Prolactin	Peptide	Stimulates milk production and secretion in breast
	Adrenocorticotropic hormone (ACTH)	Peptide	Stimulates synthesis and secretion of adrenal cortical hormones (cortisol, androgens, and aldosterone)
	Melanocyte-stimulating hormone (MSH)	Peptide	Stimulates melanin synthesis (? humans)

Hypothalamic - Hypophyseal Portal Blood Vessels



Growth depend on growth hormone but is influenced by other factors as well

- Growth hormone is essential for growth.
- Genetic determination of an individual maximum growth capacity.
- Adequate diet.
- Freedom from chronic diseases and stressful environment.
- Normal level of growth influencing hormone like thyroid hormone, insulin and sex hormones.

Growth Hormone (somatotropic hormone or somatotropin)

- Protein hormone (191 amino acids), molecular weight 21,500
- Secreted by the acidophils of the anterior pituitary,which are also known as somatotropes.

Physiological Functions of Growth Hormone

- Growth hormone promotes growth of almost all the body tissues.
- It promotes increase in size of cells, increased mitosis and differentiation of certain type of cells such as bone growth cells, muscle cells.

Effect of Growth hormone on Skeletal framework

- Increased deposition of protein by chondrocytic and osteogenic cells that causes bone growth.
- Increased rate of production of these cells.
- Conversion of chondrocytes into osteogenic cells causing deposition of new bone.
- Long bones grow in length at the epiphyseal cartilages where epiphysis at the ends of long bones are separated from shaft.
- Lengthening does not occur if epiphysis are united with the shaft.
- Growth hormone strongly stimulates osteoblasts so bones can become thicker under the influence of growth hormone throughout life.

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Growth Hormone is Potent Protein Sparer

- Growth hormone promotes protein deposition in tissues by increasing amino acid transport through cell membrane.
- Enhancement of mRNA Translation.
- Increased nuclear transcription of DNA to mRNA (over prolonged periods).
- Decreased catabolism of proteins and amino acids.

Growth Hormone Enhances Fat Utilization For Energy

- Fats are used for energy in preference to the use of carbohydrates and proteins
- Release of fatty acids from adipose tissue
- Conversion of fatty acids to acetyl-CoA
- These fatty acids are used for the production of energy by the cells, so proteins are spared.

Ketogenic Effect of Excess Growth Hormone

- The growth hormone increases free fatty acid levels in the blood which are utilized for providing energy. When excess quantities of growth hormone are present large quantities of acetoacetic acid formed by liver are released into body fluids causing Ketosis.

Growth Hormone Decreases Carbohydrate Utilization

- Growth hormone increases blood glucose levels
- Decreased glucose uptake & utilization in tissues such as muscle and fat
- Increased glucose production by the liver
- Increased Insulin secretion (causes insulin resistance)
- Reduces effect of insulin action of uptake and utilization of glucose
- Inhibits gluconeogenesis leading to **GH induced Insulin resistance** further causing increased blood glucose concentration.

- Growth Hormone Effects are Diabetogenic
- Growth hormone causes insulin resistance
- Decreased glucose utilization by the cells
- Raising blood levels of fatty acids above normal decreases the sensitivity of liver and skeletal muscle to Insulin’s effects.

Growth Hormone effect on electrolytes

- Growth hormone increases intestinal absorption of calcium.
- Growth hormone reduces excretion of sodium and potassium.

Insulin-like Growth Factors (IGFs)

- According to the traditional view GH does not directly act on its target cells to bring about its growth promoting action (increased cell division, enhanced protein synthesis, and bone growth)
- These effect are directly brought about by peptide mediators called as Somatomedins.
- Are similar to pro-insulin; produced by many tissues
- Liver produces & secretes IGF-1 in response to GH
- IGF-1 in turn stimulates cell division & growth of cartilage
- These actions are supported by IGF-2 which has more insulin-like actions.

IGF-I (somatomedin-C)

- Secretion of IGF-I before birth is independent of growth hormone but is stimulated by growth hormone after birth.
- It has pronounced growth promoting activity.
- Its concentration in plasma rises during childhood, peaks at the time of puberty and declines in the old age.
- The IGF-I receptor is very similar to Insulin receptor.

IGF-II

- IGF-II plays important role in the growth of fetus and its secretion is independent of growth hormone.

Diurnal variations in Growth hormone Secretion

- The levels of Growth hormone are low during the day
- During sleep large pulsatile bursts of Growth hormone secretion occur (specially in first two hours of deep sleep)
- Secretion of Growth hormone is under Hypothalamic control

Children display 2 periods of rapid growth

- A postnatal growth spurt during their first 2 years of life.
- A pubertal growth spurt during adolescence.

Growth hormone secretion is regulated by 2 hypophysiotropic hormones

- Growth hormone releasing hormone (GHRH) which is stimulatory.
- Growth hormone inhibiting hormone (GHIH) which is inhibitory.

Growth Hormone Stimulation:

functions in regulating growth, tissue maintenance, metabolism

GHRH from hypothalamus causes release of

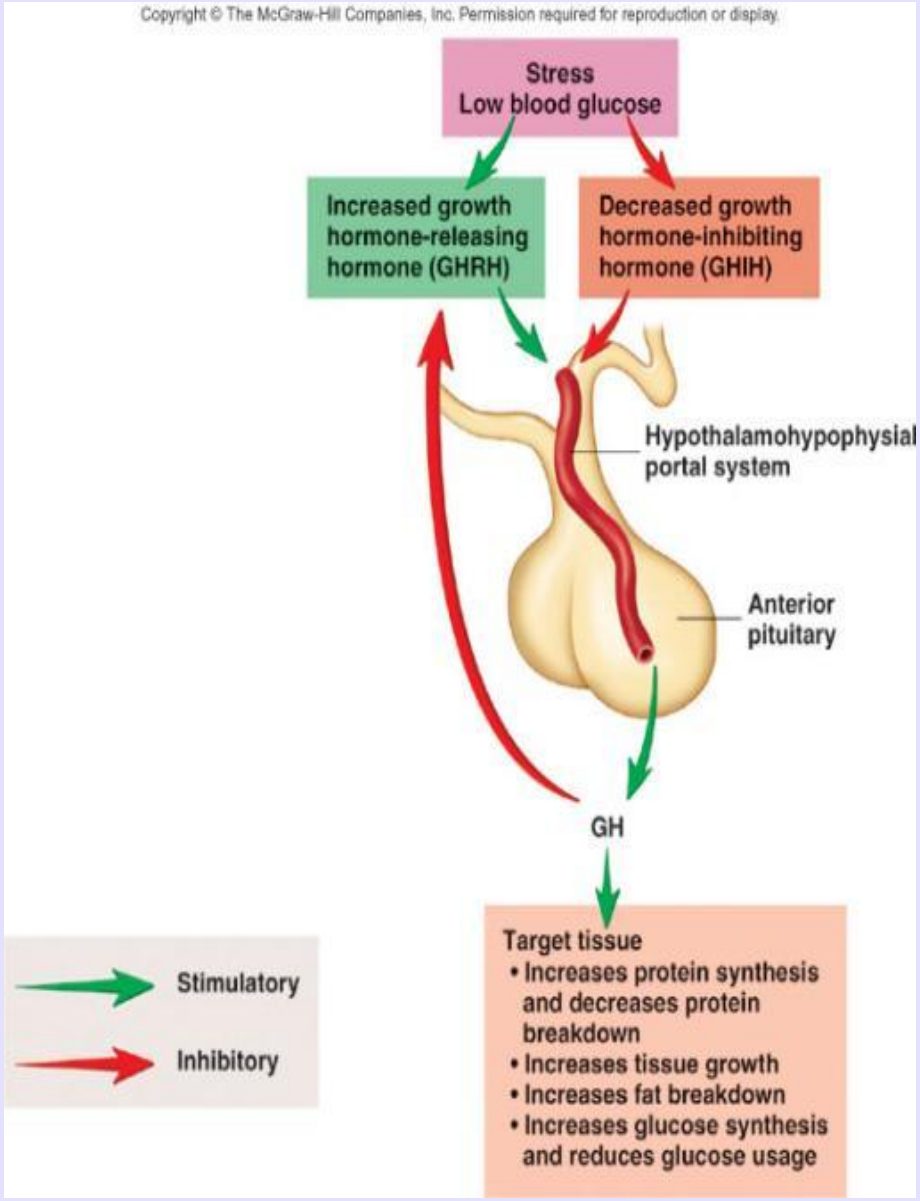
Growth hormone from anterior pituitary effects

Target tissues: most tissues of the body

- Direct effect: GH binds to receptors on cells and causes changes within the cells. Increased *lipolysis* and decreased use of glucose for energy
- Indirect effect: causes liver and skeletal muscle to produce somatomedins; e.g., insulin like growth factors (IGF's)
  - Insulin Like growth factors: bind to receptors on membranes of target cells. Stimulate growth in cartilage, bone; increased synthesis of proteins in skeletal muscle.

Regulation of GH Secretion

1. Stress and decreased glucose levels increase release of GHRH and decreased release of GHIH.
2. GHRH and GHIN travel via the hypothalamo-hypophyseal portal system to ant. pituitary
3. Increased GHRH and reduced GHIH act on AP and result in increased GH secretion.
4. GH acts on target tissues.
5. Increasing GH levels have negative feedback effect on hypothalamus.



Factors That Stimulate or Inhibit Secretion of Growth Hormone

Stimulate Growth Hormone Secretion	Inhibit Growth Hormone Secretion
Decreased blood glucose	Increased blood glucose
Decreased blood free fatty acids	Increased blood free fatty acids
Starvation or fasting, protein deficiency	Aging
Trauma, stress, excitement	Obesity
Exercise	Growth hormone inhibitory hormone (somatostatin)
Testosterone, estrogen	Growth hormone (exogenous)
Deep sleep (stages II and IV)	Somatomedins (insulin-like growth factors)
Growth hormone–releasing hormone	

Growth Hormone: Inhibition

- Hypothalamus produces growth hormone inhibiting hormone (GHIH = somatostatin)
- Inhibits production of GH by anterior pituitary.
- GHRH secretion in response to low blood glucose, stress, etc
- GHIH secretions in response to high blood glucose.
- Peak GH levels during deep sleep; levels lower at other times of day.
- Hyposecretion of GH may result in *dwarfism*
- Hypersecretion may result in *gigantism* or *acromegaly* depending on ossification of epiphyseal plates.

HYPERACTIVITY OF ANTERIOR PITUITARY

1. Gigantism

Gigantism is the pituitary disorder characterized by excess growth of the body. The subjects look like the giants with average height of about 7 to 8 feet.

- **Causes**

Gigantism is due to hypersecretion of GH in childhood or in pre-adult life before the fusion of epiphysis of bone with shaft. Hypersecretion of GH is because of tumor of acidophil cells in the anterior pituitary.Giants are hyperglycemic and they develop glycosuria and pituitary diabetes. Hyperglycemia causes constant stimulation of β-cells of islets of Langerhans in the pancreas and release of insulin. However, the overactivity of β-cells of Langerhans in pancreas leads to degeneration of these cells and deficiency of insulin and ultimately, diabetes mellitus is developed.

Gigantism

Cause

- Acidophilic tumors of anterior pituitary before puberty
- All the body tissues grow very rapidly
- If the condition develops before the union of shaft and epiphysis the person becomes giant
- There is hyperglycemia.



Acromegaly

- Acromegaly is the disorder characterized by the enlargement, thickening and broadening of bone particularly in the extremities of the body.
- **Causes**

Acromegaly is due to hypersecretion of GH in adults after the fusion of epiphysis with shaft of the bone.

Hypersecretion of GH is because of tumor of acidophil cells in the anterior pituitary.

Acromegalic or gorilla face: Face with rough features such as protrusion of supraorbital ridges, broadening of nose thickening of lips, thickening and wrinkles formation on forehead and prognathism (protrusion of lower jaw)

1. Enlargement of hands and feet
2. Kyphosis (extreme curvature of upper back thoracic spine)
3. Thickening of scalp. Scalp is also thrown into folds or wrinkles like bulldog scalp.
4. Enlargement of visceral organs such as lungs, thymus, heart, liver and spleen.
5. Hyperactivity of thyroid, parathyroid and adrenal glands.
6. Hyperglycemia and glycosuria, resulting in diabetes mellitus
7. Hypertension
8. Headache
9. Visual disturbance (bitemporal hemianopia).



Dwarfism

Dwarfism is a pituitary disorder in children characterized by the stunted growth.

- **Causes**

Reduction in GH secretion in infancy or early childhood causes dwarfism. It occurs because of the following reasons:

i. Tumor of chromophobes: It is a non-functioning tumor, which compresses and destroys the normal cells secreting GH. Most common cause for hyposecretion of GH, leading to dwarfism.

ii. Deficiency of GH-releasing hormone secreted by hypothalamus

iii. Deficiency of somatomedin C

Panhypopituitarism: In this condition, there is reduction in the secretion of all the hormones of anterior pituitary gland.



THANKS